Notes on Anytime Adaptive Data Analysis

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Abstract

There are two approaches to handling adaptivity in statistical inference. The line of work in *anytime-valid inference*, studied in statistics, allows the analyst to adaptively stop or continue collecting data based on existing results, but fixes the "query" (e.g., estimand) in advance. *Adaptive data analysis*, studied in theoretical computer science, allows the analyst to choose queries adaptively, but fixes the dataset size n in advance, and enables accurate answers to a near-quadratic number of queries. Each approach protects against one axis of adaptivity while holding the other fixed. In this work, we ask: is it possible for an analyst to be adaptive in both senses—i.e., with optional stopping/continuation, and while choosing queries strategically—while preserving statistical validity?

We study a mechanism with access to a growing stream of data that must accurately answer adaptive queries, where the analyst can also adaptively choose when to stop. In this manuscript, we formalize the problem of *anytime adaptive data analysis*; show how our formalism can concretely instantiate a sequential hypothesis testing problem; and provide a simple epochwise mechanism that provides $k \asymp n^2$ queries (up to logarithmic factors).

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1 Introduction

In many real-world data science settings, the analyst has access to a stream of data that arrives over time, rather than a static, fixed dataset. For instance, for *scientific discovery*, an analyst might hope to both analyze the data already at hand and use it to guide the analysis of future data; in the *social sciences*, where data is scarce but accumulates slowly, an analyst might be especially interested in re-using previously-seen data for estimation; in *algorithmic auditing*, additional information about the algorithm's behavior may arrive as more people interact with it.

While extensive prior work has studied how an analyst might choose to strategically stop or continue data collection for a fixed query (*anytime-valid inference*; see, e.g., Waudby-Smith and Ramdas (2024)), or how she may re-analyze the same dataset multiple times (*adaptive data analysis*; e.g., Dwork et al. (2015); Bassily et al. (2016)), it is useful to understand how the two forms of adaptivity interact. That is, in a world where data arrives over time, is it possible to allow the analyst to choose queries based on prior results, while also stopping as soon as they are satisfied with their results? The key question is whether such a mechanism can preserve *validity* in a statistical sense—whether, colloquially, the analyst can be prevented from "p-hacking."

In this note, we formalize *anytime adaptive data analysis*, which we hope can be one step towards an answer in the affirmative. Two desiderata naturally arise from our formalization of this problem. First, we ask that at any stopping time τ chosen by the analyst, the mechanism has accurately answered a near-quadratic number of queries, $\widetilde{\Omega}(\tau^2)$ for fixed accuracy and failure probability, as if τ had been a predetermined sample size. On the other hand, such a mechanism must be informative enough to remain useful to the analyst. Thus, we also ask for an analyst strategy for choosing queries so that, for the final query asked by the analyst $q_\star$, its corresponding stopping time (equivalently, sample size) τ is competitive with a world where the analyst had been asking $q_\star$ from the start.

Prior work. The study of adaptive reuse of a static sample was initiated in Dwork et al. (2015), who connected differential privacy to generalization. The core insight in this work and its follow-ups (e.g., Bassily et al. (2016); Jung et al. (2020)) was that, in the worst case, randomization is necessary to prevent an adaptive analyst from overfitting to released answers, and that the structure of this randomization determines the resulting accuracy–query tradeoff. Related privacy tools control repeated interaction through adaptive privacy filters and odometers, which track the amount of "privacy budget" consumed by prior queries (Rogers et al., 2016; Whitehouse et al., 2023). Work on changing databases instead studies privacy and empirical utility under continual observation (Dwork et al., 2010; Jain et al., 2023), growing data (Cummings et al., 2018), or fully dynamic streams (Qiu and Yi, 2025). Most closely related to our work is Marchant and Rubinstein (2025), who establish ADA guarantees when queries and their timing are adaptive but the growing-data interaction has a fixed horizon.

Recent developments in anytime-valid hypothesis testing have used e-processes as the underlying machinery that guarantees validity with optional stopping, and any "adaptivity" in such tests requires ensuring the integrity of that e-process. Waudby-Smith et al. (2025) establish log-optimality for broad classes of composite sequential tests, while Sandoval et al. (2026) allow the data source to be chosen adaptively. However, both retain a fixed composite target or global null rather than an unrestricted sequence of transcript-adaptive queries. When the end-goal is to control a family-wise error rate over many hypotheses, Bonferroni or Holm-Bonferroni corrections are typically required when the dependence structure across hypotheses is unknown (Holm, 1979). Similarly, earlier work on online multiple testing considers streams of hypotheses while controlling a global error criterion such as the false discovery rate, but does not by itself justify reusing the same data for testing and selecting hypotheses adaptively (Robertson et al., 2023; Ramdas et al., 2018; Zrnic et al., 2021).

2 Notation and preliminaries

Let $\mathbb{N} = \{1, 2, \ldots\}$ and $[m] = \{1, \ldots, m\}$. Data arrives as an infinite i.i.d. stream $X_1, X_2, \ldots \sim \mathcal{D}$ on a sample space $\mathcal{X}$, and $t \in \mathbb{N}$ denotes time, equivalently the number of observations that have arrived. For integers $u \le v$, write $X_{u:v} = (X_u, \ldots, X_v)$.

This note focuses on an analyst $\mathcal{A}$, who chooses queries, and a mechanism $\mathcal{M}$, which provides responses to the queries. At time t, the analyst $\mathcal{A}$ may interactively choose $k_t \ge 0$ queries. The j-th query and answer at time t are denoted by $q_j^{(t)} : \mathcal{X} \to [0, 1]$ and $a_j^{(t)} \in [0, 1]$, respectively. The query may depend on the transcript Π of all previously released query–answer pairs, including earlier answers at the same time. We use the convention that no error event is contributed at times for which $k_t = 0$. All probability statements are over the data stream and any randomness of the mechanism and analyst. The total number of queries answered by time t is $\sum_{s=1}^{t} k_s$. A stopping time τ is taken with respect to the filtration generated by the released transcript and the passage of time.

2.1 Fixed-sample adaptive data analysis

We begin by recalling the statistical query (SQ) model.

Definition 1 (Statistical query)**.** Let $Q = \{q : \mathcal{X} \to [0, 1]\}$ be the class of all bounded queries. Let $S = (x_1, \ldots, x_n)$, with $x_1, \ldots, x_n \overset{\text{i.i.d.}}{\sim} \mathcal{D}$, be a sample of size n from the distribution $\mathcal{D}$. For each $q \in Q$, let

$$q(S) \triangleq \frac{1}{n} \sum_{s=1}^{n} q(x_s),$$
$$q(\mathcal{D}) \triangleq \mathbb{E}_{z \sim \mathcal{D}}[q(z)].$$

A *statistical query oracle* O takes as input error tolerance $\varepsilon \in (0, 1)$, failure probability $\delta \in (0, 1)$, and a bounded query $q \in Q$, then returns answer $a \in [0, 1]$ such that

$$\mathbb{P}\left(|q(\mathcal{D}) - a| \ge \varepsilon\right) \le \delta,$$

where the probability is taken jointly over the sample and the randomness in the oracle.

A natural way to implement a SQ oracle is to draw $n = O\left(\frac{\log(1/\delta)}{\varepsilon^2}\right)$ samples from $\mathcal{D}$ and set the answer a to be the empirical mean of q over the samples. Another natural way is to add calibrated noise Z to the empirical mean and return $a = q(S) + Z$ (projected onto $[0, 1]$ if necessary), choosing the sample size and noise distribution so that both $|q(\mathcal{D}) - q(S)| \le \varepsilon/2$ and $|q(S) - a| \le \varepsilon/2$ hold simultaneously with the desired probability.

Following prior works, we formalize adaptive data analysis as the problem of answering k adaptively chosen statistical queries $q_1, \ldots, q_k \in Q$. We first recall the standard fixed-sample setting, where the number of samples and adaptive queries are fixed *a priori*, then introduce an *anytime* variant, where samples arrive in a stream, hence the number of samples and queries can grow without bound.

We start by defining the (classical) fixed-sample adaptive data analysis setting.

Definition 2 (*Fixed-sample Adaptive Data Analysis*)**.** A fixed-sample adaptive data analysis mechanism $\mathcal{M}$ works as follows. $\mathcal{M}$ takes as inputs error tolerance $\varepsilon \in (0, 1)$, failure probability $\delta \in (0, 1)$, sample $S = (x_1, \ldots, x_n)$, and query budget k. $\mathcal{M}$ then answers any k adaptive statistical queries $q_1, \ldots, q_k \in Q$, with the guarantee that

$$\mathbb{P}\left(\max_{j \in [k]} |q_j(\mathcal{D}) - a_j| \ge \varepsilon\right) \le \delta.$$

Mechanisms $\mathcal{M}$ that achieve Definition 2 are well-established in the literature (Dwork et al., 2015; Bassily et al., 2016). A standard example returns a noisy empirical answer, such as $a_j = q_j(S) + Z_j$, where the noise level Z_j is calibrated to the mechanism's stability and accuracy guarantees (Rogers et al., 2020).

For concreteness, suppose the fixed-sample mechanism can answer k queries whenever

$$n \geq C \frac{\sqrt{k} \log(1/\delta)}{\varepsilon^2} \tag{1}$$

for a universal constant $C > 0$. Equivalently, given n samples, it can answer

$$k \leq \frac{\varepsilon^4 n^2}{C^2 \log^2(1/\delta)}.$$

Thus, for fixed ε and δ, ADA can answer a quadratic number of adaptive queries up to logarithmic factors. In contrast, if we used sample splitting, i.e., an independent subsample of size $O\left(\frac{\log(k/\delta)}{\varepsilon^2}\right)$ for each adaptive query, the total sample complexity would grow linearly with k. Hence, ADA quadratically improves the dependence of the number of adaptive queries on the available sample size in this regime.

In the ADA literature, the ability to answer statistical queries is typically thought of as inherently desirable. The following example illustrates one reason to use the abstract framework of adaptive data analysis with statistical queries: ADA offers a path to "adaptive" hypothesis testing that avoids a union bound over the entire query class.

Example 1 (*Fixed-sample Adaptive Hypothesis Test*)**.** Suppose that we wish to take binary action, denoted by 1, if there exists some bounded function $q \in Q$ whose population mean under the distribution $\mathcal{D}$ is greater than $\gamma > 0$. For example, we can think of q a representation of treatment effect, and γ as the threshold at which a treatment effect becomes non-negligible. Instead of committing a priori to a set of treatment effects that we care about, suppose we wish to adaptively select q and test whether its population mean exceeds our threshold. To do this, we can repeatedly call an adaptive data analysis mechanism (with error tolerance ε and failure probability δ) with the queries $q_1, \ldots, q_k$, and receive the answers $a_1, \ldots, a_k$. Making a binary decision with the following rule,

$$\mathbb{1}\left(a_j \geq \gamma + \varepsilon\right),$$

guarantees that the probability of any false rejection is controlled because

$$\begin{aligned}\left\{\exists j \in [k] : a_j \geq \gamma + \varepsilon \text{ and } q_j(\mathcal{D}) \leq \gamma\right\} \\ \subseteq \left\{\max_{j \in [k]} |q_j(\mathcal{D}) - a_j| \geq \varepsilon\right\}.\end{aligned}$$

Therefore,

$$\mathbb{P}\left(\exists j \in [k] : a_j \geq \gamma + \varepsilon \text{ and } q_j(\mathcal{D}) \leq \gamma\right) \leq \delta.$$

Under (1), the mechanism can answer a near-quadratic number of adaptive queries using n samples, without a dependence on the size of Q. In contrast, sample splitting allows only a near-linear number of adaptive queries using n samples.

3 Anytime adaptive data analysis

In this section, we introduce our formalization of *anytime adaptive data analysis*. In this setting, samples arrive in a stream (still i.i.d. with respect to $\mathcal{D}$), and we ask that the number of adaptive queries answered grow with the number of samples observed.

Definition 3 (Anytime Adaptive Data Analysis)**.** An *anytime adaptive data analysis* mechanism $\mathcal{M}$ works as follows. $\mathcal{M}$ takes as inputs error tolerance $\varepsilon \in (0,1)$ and failure probability $\delta \in (0,1)$. Data arrives as an infinite stream $X_1, X_2, \ldots$, indexed by t. At time t, the mechanism answers $k_t \geq 0$ adaptive statistical queries $q_1^{(t)}, \ldots, q_{k_t}^{(t)} \in Q$, with the guarantee that

$$\mathbb{P}\left(\exists t \geq 1,\, \exists j \in [k_t] : |q_j^{(t)}(\mathcal{D}) - a_j^{(t)}| \geq \varepsilon\right) \leq \delta.$$

In the following example, we show why a mechanism that satisfies Definition 3 is useful for the anytime hypothesis testing setting.

Example 2 (Anytime Adaptive Hypothesis Test)**.** The typical "anytime-valid" hypothesis test at level α asks for a guarantee of the form $\mathbb{P}\left(\exists t \geq 1,\, \mathcal{H}_0 \text{ erroneously rejected at } t\right) \leq \alpha$. Consider the class of null hypotheses restricted to *thresholded means*, i.e.,

$$\mathcal{H}_0 : q(\mathcal{D}) \leq \gamma \qquad \mathcal{H}_1 : q(\mathcal{D}) > \gamma.$$

For every query $q_j^{(t)}$ and answer $a_j^{(t)}$, the false-rejection event satisfies

$$\left\{a_j^{(t)} \geq \gamma + \varepsilon \text{ and } q_j^{(t)}(\mathcal{D}) \leq \gamma\right\} \subseteq \left\{|a_j^{(t)} - q_j^{(t)}(\mathcal{D})| \geq \varepsilon\right\}.$$

Thus, a mechanism satisfying Definition 3 immediately achieves level-δ Type I error control:

$$\begin{aligned}\mathbb{P}\left(\exists t \geq 1,\, \exists j \in [k_t] : a_j^{(t)} \geq \gamma + \varepsilon \text{ and } q_j^{(t)}(\mathcal{D}) \leq \gamma\right) \\ \leq \mathbb{P}\left(\exists t \geq 1,\, \exists j \in [k_t] : |q_j^{(t)}(\mathcal{D}) - a_j^{(t)}| \geq \varepsilon\right) \leq \delta.\end{aligned}$$

3.1 What are the benefits of adaptivity?

The benefit of adaptive query selection depends on whether earlier answers merely choose among a fixed, unstructured list or instead help construct the final query. Appendix A makes this comparison precise in two examples. Both use the oracle benchmark $n_0(\Delta) \asymp 1/\Delta^2$, the sample size needed to test a known subgroup with gap Δ, and measure the additional cost of searching over K possible subgroups.

In the unstructured scan of Example 3, generic ADA must be prepared to answer as many as K adaptively selected queries. Its additional multiplicity cost is therefore $\widetilde{O}(\sqrt{K}/\Delta^2)$, which is worse in K than the $O(\log K/\Delta^2)$ cost of Bonferroni-calibrated per-hypothesis e-values. Changing the order of the scan cannot improve this worst-case comparison: answers about null hypotheses need not reveal the location of the signal.

The picture changes when previous answers can eliminate a constant fraction of the candidates and thereby construct the final audit query. In the hierarchical-group and adaptive-rectangle searches of Example 4, only $T = O(\log K)$ queries are needed. The same generic ADA rate then gives additional multiplicity cost $\widetilde{O}(\sqrt{\log K}/\Delta^2)$, improving on an unstructured Bonferroni scan. This gain comes from the routing margin or analogous search structure, not from hierarchy or geometry alone.

The comparison can be summarized as follows:

	Bonferroni e-values	generic ADA
unstructured scan	$O\left(\frac{\log K}{\Delta^2}\right)$	$\widetilde{O}\left(\frac{\sqrt{K}}{\Delta^2}\right)$
adaptive construction	$O\left(\frac{\log K}{\Delta^2}\right)$	$\widetilde{O}\left(\frac{\sqrt{\log K}}{\Delta^2}\right)$.

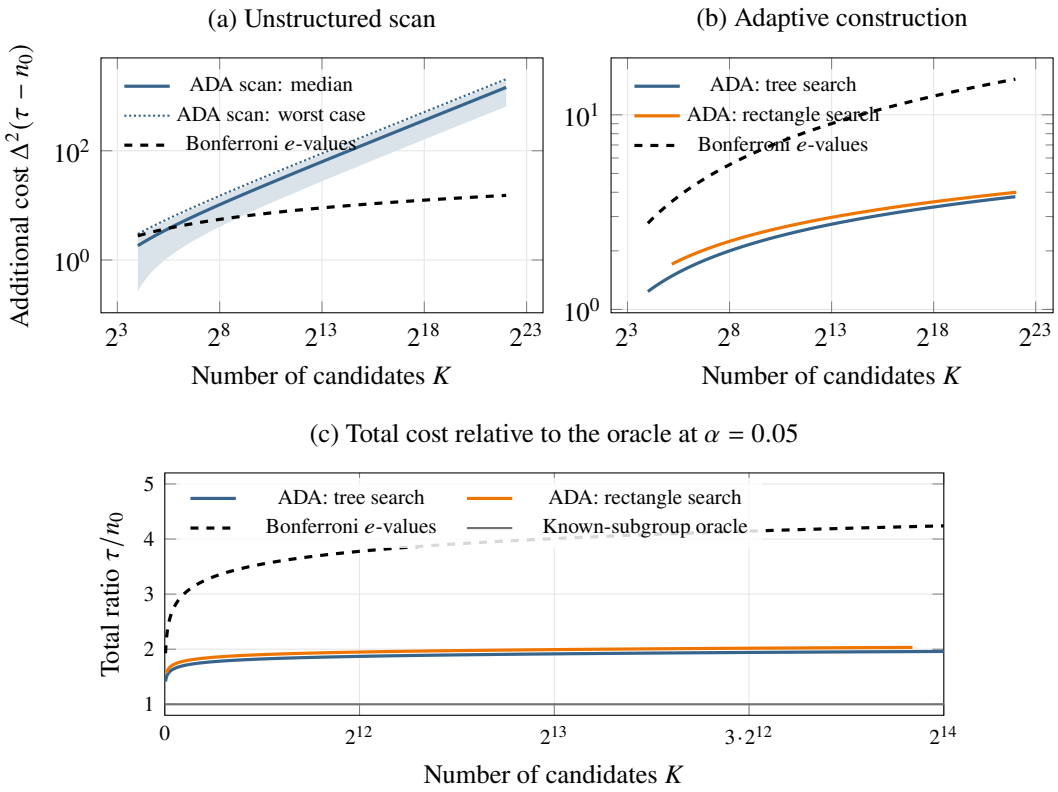


Figure 1: Multiplicity cost for an unstructured scan and adaptive construction. Panels (a) and (b) show the additional effective sample size beyond the known-subgroup benchmark, in units of $1/\Delta^2$. Panel (c) sets the baseline testing cost at $\alpha = 0.05$ and plots the multiplicative total-cost ratio τ/n_0 relative to the oracle benchmark (so a value of 2 means twice the oracle sample size, rather than an additive overhead of 2). Full examples and plotting details appear in Appendix A.

Figure 1 makes the selection–construction distinction visible. In panel (a), the generic ADA cost grows as $\sqrt{K}$ and eventually exceeds the logarithmic Bonferroni cost. In panel (b), a successful auditor converts each answer into a constant-factor reduction of the candidate set, leaving the ADA mechanism responsible for only order $\log K$ queries. Panel (c) shows that, once the unavoidable testing cost is restored, strategic construction remains much closer to the known-subgroup benchmark over the displayed range.

3.2 An initial algorithmic attempt

In Algorithm 1, we provide an approach to achieving anytime adaptive data analysis, as in Definition 3, that uses a *black-box* mechanism $\mathcal{M}$ (without requiring new algorithmic development for the mechanism itself). At a high level, Algorithm 1 divides the time horizon into exponentially increasing time periods, and applies the black-box fixed-sample ADA mechanism $\mathcal{M}$ separately to each completed period.

Theorem 1 (Doubling trick)**.** *Suppose the fixed-sample ADA mechanism $\mathcal{M}$ satisfies Definition 2 whenever* (1) *holds. Then Algorithm 1 satisfies anytime ADA (Definition 3). Moreover, for fixed ε and δ, the total number of queries answered by time t is $\widetilde{\Theta}(t^2)$, where the suppressed factor is logarithmic in the epoch index (equivalently, doubly logarithmic in t).*

Proof of Theorem 1. The i-th release occurs at time $2^i - 1$ and uses the completed sample $S_i = X_{2^{i-1}:(2^i-1)}$,

Algorithm 1 Doubling trick, fixed ε

Require: Error tolerance $\varepsilon \in (0,1)$, failure probability $\delta \in (0,1)$, fixed-sample ADA mechanism $\mathcal{M}$ satisfying (1)

Initialize transcript Π to be an empty list and $i = 1$
for $t = 1, 2, \ldots$ **do**
  Observe $X_t \sim \mathcal{D}$ and set $k_t \leftarrow 0$
  if $t = 2^i - 1$ **then**
    Set $\delta_i \leftarrow \frac{6}{\pi^2} \cdot \frac{\delta}{i^2}$ and $S_i \leftarrow X_{2^{i-1}:(2^i-1)}$
    Set $k_t \leftarrow \left\lfloor \frac{\varepsilon^4 (2^{i-1})^2}{C^2 \log^2(1/\delta_i)} \right\rfloor$
    Initialize $\mathcal{M}_i \leftarrow \mathcal{M}[\varepsilon, \delta_i, S_i, k_t]$
    for $j = 1, 2, \ldots, k_t$ **do**
      Query $q_j^{(t)} \sim \mathcal{A}(\Pi)$
      Answer $a_j^{(t)} \sim \mathcal{M}_i(q_j^{(t)}, \Pi)$
      Update $\Pi \leftarrow \Pi \oplus (q_j^{(t)}, a_j^{(t)})$
    end for
    $i \leftarrow i + 1$
  end if
end for

which has size 2^{i-1}. By the definition of k_{2^i-1},

$$2^{i-1} \geq C \frac{\sqrt{k_{2^i-1}} \log(1/\delta_i)}{\varepsilon^2},$$

so Definition 2 applies to epoch i.

Conditional on the transcript before epoch i, the new sample S_i is independent of that transcript. Therefore, the fixed-sample ADA guarantee applies conditionally even though the analyst's strategy in epoch i may depend on all previous releases. Averaging over the previous transcript and taking a union bound gives

$$\begin{aligned}
&\mathbb{P}\left(\exists t \geq 1,\, \exists j \in [k_t] : |q_j^{(t)}(\mathcal{D}) - a_j^{(t)}| \geq \varepsilon\right) \\
&\quad = \mathbb{P}\left(\exists i \geq 1,\, \exists j \in [k_{2^i-1}] : |q_j^{(2^i-1)}(\mathcal{D}) - a_j^{(2^i-1)}| \geq \varepsilon\right) \\
&\quad \leq \sum_{i=1}^{\infty} \delta_i = \sum_{i=1}^{\infty} \frac{6}{\pi^2} \frac{\delta}{i^2} = \delta.
\end{aligned}$$

Finally,

$$\log(1/\delta_i) = \log\left(\frac{\pi^2 i^2}{6\delta}\right),$$

so the epoch budgets grow geometrically as 4^{i-1} up to a factor polynomial in $\log i$. Their cumulative sum is therefore dominated, up to constant and logarithmic factors, by the most recent completed epoch. Since the most recent release time is within a constant factor of t, $\sum_{s=1}^{t} k_s = \widetilde{\Theta}(t^2)$ for fixed ε and δ. □

While this approach achieves near-quadratic asymptotic rates, several clear limitations remain. First, because of the divided time periods, the same samples are reused across queries within an epoch but are never reused across epochs, which suggests some unnecessary inefficiencies. On the other hand, the analyst must wait increasingly long intervals before she can ask her next set of queries, which would appear to dilute the benefit of optional stopping.

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A Detailed multiplicity comparisons

A.1 Full examples

Example 3 (One signal among K candidate hypotheses)**.** *Setup and oracle benchmark.* Fix K bounded queries $q_1, \dots, q_K$ and consider the hypotheses

$$H_{0,k} : q_k(\mathcal{D}) \le \gamma \qquad \text{versus} \qquad H_{1,k} : q_k(\mathcal{D}) > \gamma.$$

Suppose that there is a unique $k_\star \in [K]$ such that $q_{k_\star}(\mathcal{D}) = \gamma + \Delta$, while $q_k(\mathcal{D}) \le \gamma$ for every $k \ne k_\star$. For fixed error probability, the information-theoretic oracle benchmark in which $k_\star$ is known in advance requires the effective sample size

$$n_0(\Delta) \asymp \frac{1}{\Delta^2} \tag{2}$$

in the usual bounded-mean model. Here and in Example 4 we hold the testing level fixed in order to isolate the cost of multiplicity; this deliberately suppresses mechanism-dependent $\log(1/\delta)$ factors. We call the K-dependent term beyond $n_0(\Delta)$ the *additional multiplicity cost*.

Bonferroni/e-value benchmark. For comparison, suppose that an e-process $E_{k,t}$ is run for each of the K fixed hypotheses. Rejecting $H_{0,k}$ when $E_{k,t} \ge K/\delta$ controls the family-wise error rate at level δ by Ville's inequality and a union bound. For a standard bounded-mean e-process whose log growth under the alternative is $\Theta(t\Delta^2)$, its stopping time is therefore

$$\tau_e = O\left(\frac{\log(1/\delta) + \log K}{\Delta^2}\right).$$

Thus Bonferroni calibration adds $\log K/\Delta^2$ samples to the oracle benchmark (equivalently, an additive $\log K$ in the evidence boundary) (Holm, 1979; Waudby-Smith and Ramdas, 2024).

ADA strategy. Now consider an analyst interacting with Algorithm 1. To separate the mechanism's error tolerance from the effect size, denote the former by η. On the simultaneous-accuracy event, querying $q_{k_\star}$ gives

$$a_{k_\star} \ge \gamma + \Delta - \eta.$$

Consequently, the rejection rule in Example 2, which rejects when $a_k \ge \gamma + \eta$, is guaranteed to reject the non-null once it is queried provided $\Delta > 2\eta$. In particular, the weaker condition $\Delta > \eta$ by itself is not sufficient for guaranteed power under the abstract ADA accuracy guarantee. If Δ is known, one may choose any $\eta < c\Delta$ for a constant $c < 1/2$.

With no side information distinguishing the K hypotheses, a natural strategy is to query the hypotheses in a uniformly random order, always using an available query slot for a hypothesis that has not yet been queried. Repeating a query at the same tolerance cannot improve worst-case power under the black-box guarantee. Let

$$B_i = \sum_{\ell=1}^{i} \left\lfloor \frac{\eta^4 (2^{\ell-1})^2}{C^2 \log^2(1/\delta_\ell)} \right\rfloor$$

be the cumulative number of available query slots through epoch i. If the signal's rank in the random ordering is R, the analyst rejects at the first epoch for which $B_i \ge R$, except on the probability-δ accuracy failure event. Under the random ordering, R is uniform on $[K]$ for every fixed location $k_\star$, so

$$\mathbb{P}(R \le B_i) = \min\{B_i/K, 1\}, \qquad \mathbb{E}[R] = (K+1)/2;$$

in the worst case, rejection is guaranteed only once $B_i \geq K$. Because the most recent epoch dominates B_i, this occurs when, up to constants,

$$2^{i-1} \gtrsim \frac{C\sqrt{K}\log(1/\delta_i)}{\eta^2}.$$

Choosing $\eta = c\Delta$ therefore gives

$$\begin{aligned}\tau_{\mathrm{ADA}} &= \widetilde{O}\left(\frac{\sqrt{K}}{\Delta^2}\right) \\ &= \widetilde{O}\left(n_0(\Delta) + \frac{\sqrt{K}-1}{\Delta^2}\right)\end{aligned}$$

for fixed δ, where the additional logarithmic factor comes from the epochwise allocation of δ.

Additional multiplicity cost. At the level of dependence on K, the two procedures can now be compared relative to the same known-subgroup benchmark:

$$\begin{aligned}\mathrm{MultCost}_{e\text{-Bonf}} &= O\left(\frac{\log K}{\Delta^2}\right), \\ \mathrm{MultCost}_{\mathrm{ADA,scan}} &= \widetilde{O}\left(\frac{\sqrt{K}-1}{\Delta^2}\right).\end{aligned}$$

Thus generic ADA applied to a finite, unstructured, pre-specified family is actually worse in K than Bonferroni-calibrated per-hypothesis e-values. The value of ADA must instead come from settings in which earlier answers are used to construct the final query, or the effective query class cannot be enumerated in advance.

Why the scan cannot improve without structure. The same calculation explains why answers released early must be inaccurate if the analyst insists on querying all K hypotheses in each epoch. Algorithm 1 itself holds the tolerance fixed, so it cannot release such a K-query batch until its budget reaches K; allowing early batches requires a varying-tolerance extension. For an epoch containing $n_i = 2^{i-1}$ observations, rearranging (1) shows that the error radius for a batch of K queries can be no smaller than

$$\eta_i \geq \left(\frac{C\sqrt{K}\log(1/\delta_i)}{n_i}\right)^{1/2} = \frac{\sqrt{C\log(1/\delta_i)}K^{1/4}}{\sqrt{n_i}}.$$

These simultaneous answers cannot guarantee rejection until $\eta_i < \Delta/2$, which again requires

$$n_i > \frac{4C\sqrt{K}\log(1/\delta_i)}{\Delta^2}.$$

Among strategies restricted to the individual queries $q_1, \ldots, q_K$, this no-repeat scan is minimax-optimal in the unstructured worst case: until $q_{k_\star}$ is queried at sufficient accuracy, answers concerning other hypotheses need not reveal anything about its location. Structured relationships among the q_k, or permission to ask informative aggregate queries, could lead to a different strategy and are not captured by this example.

Example 4 (Constructing the final query by adaptive search)**.** *Setup and oracle benchmark.* Example 3 treats the K candidate queries as an unstructured list. In contrast, suppose that earlier answers allow the analyst to eliminate a constant fraction of the candidates and construct the final query $q_\star$. Once $q_\star$ is known, suppose that $q_\star(\mathcal{D}) = \gamma + \Delta$. Testing this known subgroup again has the oracle effective sample size $n_0(\Delta) \asymp 1/\Delta^2$ from (2).

Bonferroni/e-value benchmark. If the structure is ignored and all K candidate subgroups are instead tested with Bonferroni-calibrated per-hypothesis e-values, then

$$\tau_e = O\left(n_0(\Delta) + \frac{\log K}{\Delta^2}\right).$$

Its additional multiplicity cost is therefore $O(\log K/\Delta^2)$, exactly as in Example 3.

ADA strategy: hierarchical groups. First, let $K = 2^m$ candidate subgroups be the leaves of a known binary tree, with one violating leaf $G_\star$. Suppose that at every internal node v there is a bounded routing query r_v satisfying

$$\begin{aligned} r_v(\mathcal{D}) &\geq \frac{1}{2} + \Delta && \text{if } G_\star \text{ is in the left subtree of } v, \\ r_v(\mathcal{D}) &\leq \frac{1}{2} - \Delta && \text{if } G_\star \text{ is in the right subtree of } v. \end{aligned} \tag{3}$$

Starting at the root, the analyst asks r_v and follows the left child when the answer is at least $1/2$, and the right child otherwise. With ADA error tolerance $\eta = \Delta/4$, every decision is correct on the simultaneous accuracy event. After $m = \log_2 K$ answers the analyst has identified $G_\star$, constructs its audit query $q_\star = q_{G_\star}$, and asks this as one final query.

ADA strategy: adaptive rectangles. Adaptive rectangle search gives a less explicitly hierarchical example. Let the candidate groups be the axis-aligned rectangles on a J-point grid in d dimensions, so

$$K = \left(\frac{J(J+1)}{2}\right)^d = \Theta(J^{2d}).$$

Suppose the population audit score has a unique maximizing rectangle $R_\star$, and has the following robust search property: for each unresolved endpoint, the analyst can form a bounded comparison query whose population mean is at least $1/2 + \Delta$ or at most $1/2 - \Delta$ according to which half of the current search interval contains the corresponding endpoint of $R_\star$. A separable, uniformly separated unimodal score in the $2d$ rectangle endpoints is one simple setting with this property. The analyst uses each answer to halve one endpoint's search interval. After at most

$$2d\lceil \log_2 J \rceil = O(\log K)$$

comparison queries, it has constructed $q_\star = q_{R_\star}$ and can ask the final audit query.

Additional multiplicity cost. In either construction, including the final audit query, the total number of adaptively selected queries is $T = 1 + O(\log K)$. The usual rule rejecting when the final answer is at least $\gamma + \eta$ succeeds for $\eta = \Delta/4$. Substituting this query count and tolerance into (1), and applying the epochwise construction, gives the stopping-time bound

$$\begin{aligned} \tau_{\mathrm{ADA,search}} &= \widetilde{O}\left(\frac{\sqrt{1 + \log K}}{\Delta^2}\right) \\ &= \widetilde{O}\left(n_0(\Delta) + \frac{\sqrt{1 + \log K} - 1}{\Delta^2}\right) \end{aligned}$$

for fixed failure probability. The additional multiplicity costs are thus

$$\mathrm{MultCost}_{e\text{-Bonf}} = O\left(\frac{\log K}{\Delta^2}\right),$$
$$\mathrm{MultCost}_{\mathrm{ADA,search}} = \widetilde{O}\left(\frac{\sqrt{1+\log K}-1}{\Delta^2}\right) = \widetilde{O}\left(\frac{\sqrt{\log K}}{\Delta^2}\right).$$

Unlike the unstructured scan in Example 3, generic ADA now has the more favorable dependence on K: earlier answers are used to construct, rather than merely select, $q_\star$.

The routing margin in (3), or its rectangle analogue, is the substantive structural assumption. Hierarchical labels or rectangle geometry alone are insufficient: if an aggregate query dilutes the signal as its region grows, an answer need not eliminate half of the candidates. This assumption is absent from Example 3, where queries about the null hypotheses reveal nothing about the location of the signal. The comparison above is also specifically to Bonferroni over the enumerated groups; a structure-aware sequential procedure may be able to exploit the same hierarchy or geometric complexity.

A.2 Construction of Figure 1

Figure 1 is a rate illustration rather than an empirical evaluation of a particular ADA implementation: the mechanism in this note is specified through the black-box rate (1). We normalize constants so that one query costs the known-subgroup benchmark $n_0(\Delta) = 1/\Delta^2$. Accommodating T adaptive queries then has effective sample size $\sqrt{T}/\Delta^2$, and hence additional cost $\sqrt{T} - 1$ in units of $1/\Delta^2$. The corresponding additional Bonferroni cost is $\log K$ in the same units.

For the unstructured scan in Example 3, panel (a) approximates the signal's rank R as continuous uniform on $[0, K]$ and sets $T = R$. It reports the median and 10th–90th percentile curves, together with the worst-case $T = K$ curve. For the structured searches in Example 4, panel (b) uses $T = 1 + \log_2 K$ for the binary tree. For rectangles it fixes $d = 2$, uses $K = \{J(J+1)/2\}^2$, and sets $T = 1 + 4\log_2 J$. These continuous approximations suppress integer and rounding artifacts so that the curves emphasize the rate comparison.

Panel (c) restores the testing-level dependence that the first two panels suppress. In the same rate-calibrated normalization, write

$$n_0(\Delta, \alpha) = \frac{\log(1/\alpha)}{\Delta^2}, \qquad \tau = \frac{\log(1/\alpha) + c(K)}{\Delta^2},$$

where $c(K)$ is the additional multiplicity coefficient from panels (a) and (b). At $\alpha = 0.05$, panel (c) plots $\tau/n_0 = 1 + c(K)/\log(20)$ for the two strategic-search methods and Bonferroni, with the known-subgroup oracle shown at one. This normalization isolates the dependence on K; it does not assert that suppressed constants or mechanism-specific confidence factors are identical across procedures. All curves and plotting data are generated by `simulate_examples.py`.